

Giugno 2025

$$X \sim N(0, 1) \quad \rho_{xy} = \frac{1}{2} \quad Z = X - \frac{Y}{2}$$

$$Y \sim N(1, 4)$$

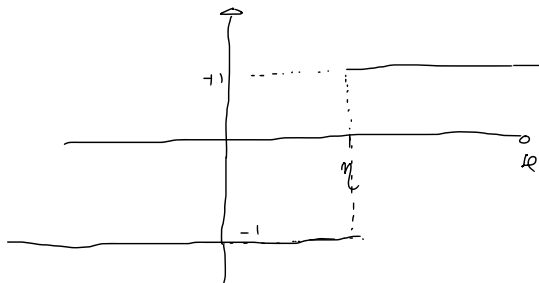
$$\mu_z = E[Z] = E[X] - \frac{1}{2} E[Y] = 0 - \frac{1}{2} \times 1 = -\frac{1}{2}$$

$$\sigma_z^2 = E[(Z - \mu_z)^2] = E\left[\left(X - \frac{Y}{2} - \left(\mu_x - \frac{1}{2}\mu_y\right)\right)^2\right] = E\left[\left[\left(X - \mu_x\right) - \frac{1}{2}\left(Y - E[Y]\right)\right]^2\right]$$

$$= E\left[\left(X - \mu_x\right)^2\right] + \frac{1}{4} E\left[\left(Y - \mu_y\right)^2\right] - E\left[\left(X - \mu_x\right)\left(Y - \mu_y\right)\right] =$$

$$= \sigma_x^2 + \frac{1}{4} \sigma_y^2 - \text{Cov}(X, Y) = \sigma_x^2 + \frac{1}{4} \sigma_y^2 - \rho_{xy} \sigma_x \sigma_y = 1 + \frac{1}{4} \times 4 - \frac{1}{2} \sqrt{1} \sqrt{4} =$$

$$= 1 + 1 - 1 = 1$$



$$V = 1 \iff Z > \eta$$

$$\downarrow$$

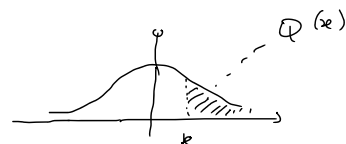
$$N\left(-\frac{1}{2}, 1\right)$$

$$P[V=1] = P[Z > \eta] = Q\left[\frac{\eta + 1/2}{1}\right] = Q\left[\eta + \frac{1}{2}\right]$$

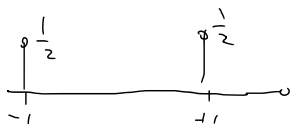
$$Q(x) = \frac{1}{2} \iff x = 0$$

$$P[V=-1] = 1 - P[V=1] = 1 - Q\left[\eta + \frac{1}{2}\right]$$

$$P\left[N(\mu, \sigma^2) > u\right] = Q\left[\frac{u - \mu}{\sigma}\right]$$



$$Q\left[\eta + \frac{1}{2}\right] = \frac{1}{2} \iff \eta + \frac{1}{2} = 0 \quad \eta = -\frac{1}{2}$$



$$E[V] = 1 \times \frac{1}{2} + (-1) \times \frac{1}{2} = 0$$

$$\text{VAR}[V] = E[V^2] = (1)^2 \times \frac{1}{2} + (-1)^2 \times \frac{1}{2} = 1$$

Settembre 2025

$$X \sim N(1, 4) \quad Z = 3X - 2Y$$

$$Y \sim N(-1, 9)$$

$$\mu_z = 3 \times 1 - 2 \times (-1) = 5$$

$$\sigma_z^2 = (3)^2 \sigma_x^2 + (-2)^2 \sigma_y^2 + 2(3)(-2) \text{Cov}[X, Y] = 9 \times 4 + 4 \times 9 - 12 \times \frac{\sqrt{4} \sqrt{9}}{3}$$

$$= 72 - 12 = 60$$

1) PDF di Z è funzione di $\rho_{xy} = \rho$

$$\sigma_z^2 = E[(Z - \mu_z)^2] = E\left[\left(3(X - \mu_x) - 2(Y - \mu_y)\right)^2\right] = (3)^2 \sigma_x^2 + (-2)^2 \sigma_y^2 + 2(3)(-2) \text{Cov}[X, Y]$$

$$\sigma_z^2 = \frac{1}{2} + \frac{1}{2} = 1 \quad \Rightarrow \quad \sigma_z = 1$$

$$f_z(z) = \frac{1}{\sqrt{2\pi} \sigma_z \sqrt{1-\rho}} \exp\left[-\frac{1}{2} \frac{1}{\sigma_z^2(1-\rho)} (z-5)^2\right]$$

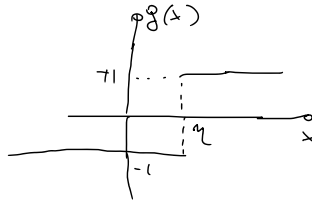
$$= \frac{1}{\sqrt{2\pi} \cdot 1 \cdot \sqrt{1-\rho}} \exp\left[-\frac{1}{2} \frac{1}{1 \cdot (1-\rho)} (z-5)^2\right]$$

$$= \frac{1}{\sqrt{2\pi} \sqrt{1-\rho}} \exp\left[-\frac{1}{2} \frac{1}{1-\rho} (z-5)^2\right]$$

$$\sigma_z = 1 \quad \Rightarrow \quad \sigma_z^2 = 1$$

$$z) \in [U] \quad \text{e} \quad \text{VAR}[U]$$

$$V=1 \quad \Leftrightarrow \quad z > \eta$$



$$P[U=1] = P[z > \eta] = Q\left[\frac{\eta - \mu_z}{\sigma_z}\right] = Q\left[\frac{\eta - 5}{\sigma \sqrt{2(1-\rho)}}\right]$$

$$P[U=-1] = 1 - P[U=1] = 1 - Q\left[\frac{\eta - 5}{\sigma \sqrt{2(1-\rho)}}\right]$$

$$E[U] = 1 \cdot P[U=1] + (-1) \cdot P[U=-1] = Q\left[\frac{\eta - 5}{\sigma \sqrt{2(1-\rho)}}\right] - \left[1 - Q\left[\frac{\eta - 5}{\sigma \sqrt{2(1-\rho)}}\right]\right] = 2Q\left[\frac{\eta - 5}{\sigma \sqrt{2(1-\rho)}}\right] - 1$$

$$E[U^2] = (1)^2 P[U=1] + (-1)^2 P[U=-1] = Q(\cdot) + [1 - Q(\cdot)] = 1$$

$$\text{VAR}[U] = E[U^2] - E^2[U] = 1 - \left(2Q(\cdot) - 1\right)^2 = 1 - \left[4Q^2(\cdot) - 4Q(\cdot) + 1\right]$$

$$= 4Q(\cdot) - 4Q^2(\cdot) = 4Q(\cdot) [1 - Q(\cdot)]$$

Luglio 2025

$X \sim \text{esponenziale di media 1}$

$B \in \{-1, 1\}$, indep. da X con $P[B=1] = \frac{1}{2}$

$$Z = BX$$

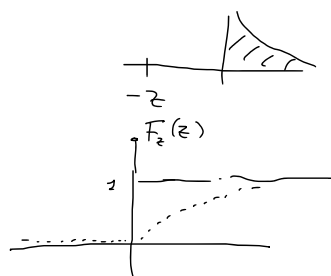
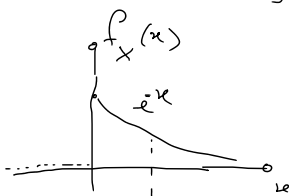
1) PDF e CDF di Z

$$F_z(z) = P[Z \leq z] = P[BX \leq z] = P[BX \leq z | B=1] P[B=1] + P[BX \leq z | B=-1] P[B=-1]$$

$$= \frac{1}{2} \left[P[X \leq z | B=1] + P[-X \leq z | B=-1] \right] = \frac{1}{2} \left[F_X(z) + P[X \geq -z] \right] =$$

$$= \frac{1}{2} \left[F_X(z) + 1 - P[X < -z] \right] = \frac{1}{2} \left[F_X(z) + 1 - F_X(-z) \right]$$

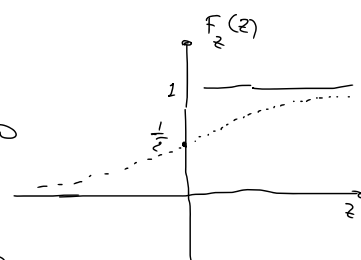
$$= \frac{1}{2} \left[1 + F_X(z) - F_X(-z) \right]$$



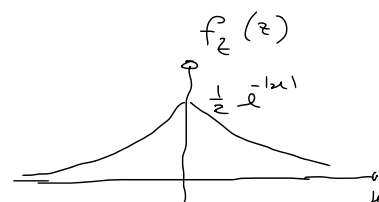
$$F_z(z) = [1 - e^{-z}] u(z)$$

$$\int_0^x e^t dt = 1 - e^{-x}$$

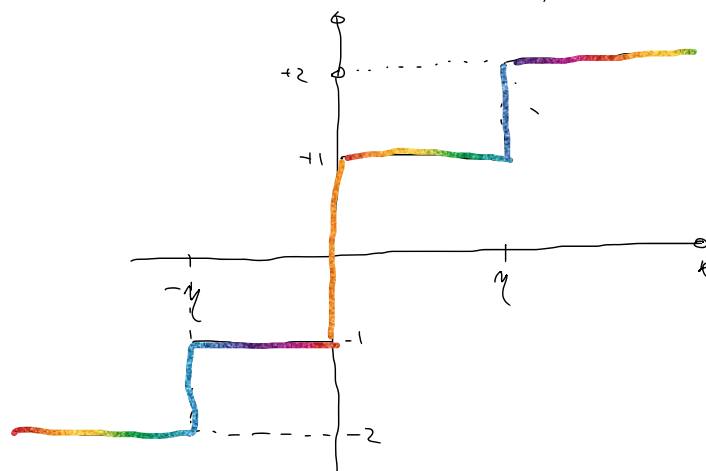
$$F_z(z) = \begin{cases} \frac{1}{2} + \frac{1}{2} F_X(z) = \frac{1}{2} + \frac{1}{2} (1 - e^{-z}) = 1 - \frac{1}{2} e^{-z} & z > 0 \\ \frac{1}{2} - \frac{1}{2} F_X(-z) = \frac{1}{2} - \frac{1}{2} (1 - e^z) = \frac{1}{2} e^z & z < 0 \end{cases}$$



$$f_z(z) = \frac{d}{dz} F_z(z) = \begin{cases} \frac{1}{2} e^{-z} & z > 0 \\ \frac{1}{2} e^z & z < 0 \end{cases} = \frac{1}{2} e^{-|z|}$$



$$V = \begin{cases} -2 & z \leq -\eta \\ -1 & -\eta < z \leq 0 \\ +1 & 0 < z \leq \eta \\ +2 & z > \eta \end{cases}$$



$$P[V=2] = P[Z > \eta] = 1 - P[Z \leq \eta] = 1 - F_Z(\eta) = 1 - \left(1 - \frac{1}{2}e^{-\eta}\right) = \frac{1}{2}e^{-\eta}$$

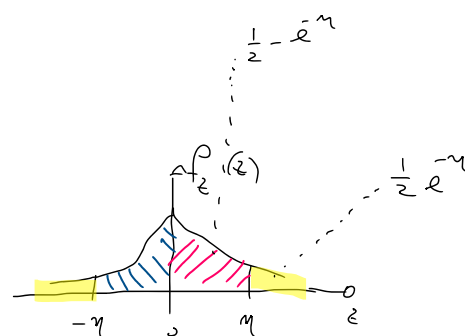
$$P[V=1] = P[0 < Z < \eta] = \frac{1}{2} - \frac{1}{2} e^{-\eta} = \frac{1}{2} (1 - e^{-\eta})$$

$$Z > 0 \iff (0 < Z < n) \cup (Z \geq n)$$

$$P(Z > 0) = P[0 < Z < n] + P[Z > n]$$

$$\frac{1}{2} = P[0 < Z < \infty) + \frac{1}{2} e^{-\eta}$$

$$P[V=-1] = P[V=1] \quad P[V=-2] = P[V=2]$$



$$V = -1 \quad \text{---} \quad -\eta < z < 0$$

$$- \eta < z < 0$$

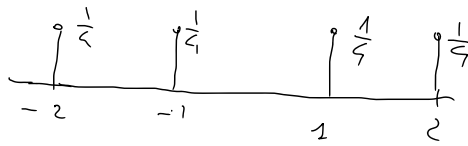
$$p[-\eta < z < 0] = \int_{-\eta}^0 f_z(z) dz \stackrel{z' = -z}{=} \int_0^{\eta} f_z(-z') dz' = \int_0^{\eta} f_z(z') dz' = P[0 < z < \eta]$$

$$V = -2 \leq 0 \quad z \leq -2$$

3) $\frac{1}{\sqrt{2}} \begin{pmatrix} -\eta \\ -\eta \end{pmatrix}$

$$V = -2 \Leftrightarrow z \leq -\eta$$

$$3) \quad \frac{1}{2} e^{-\eta} = \frac{1}{4} \quad e^{-\eta} = \frac{1}{2} \quad -\eta = \ln\left(\frac{1}{2}\right) \quad \eta = -\ln\left(\frac{1}{2}\right) = \ln(2)$$



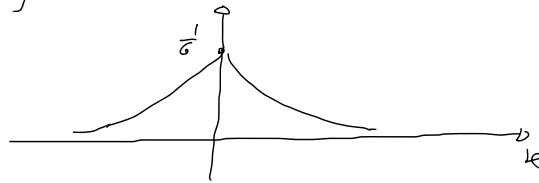
$$E[V] = 1 \times \frac{1}{2} + 2 \times \frac{1}{4} + (-1) \times \frac{1}{2} + (-2) \times \frac{1}{4} = 0$$

$$\text{VAR}[V] = E[V^2] = (1)^2 \times \frac{1}{2} + (2)^2 \times \frac{1}{4} + (-1)^2 \times \frac{1}{2} + (-2)^2 \times \frac{1}{4} =$$

$$= 2 \left\{ \frac{1}{2} + 1 \right\} = 2 \frac{5}{4} = \frac{5}{2}$$

Dicembre 2025

$$f_X(x) = \frac{1}{6} \exp\left[-\frac{|x|}{3}\right] \quad \forall x \in \mathbb{R}$$



$\Rightarrow$ CDF, $E[X]$ e $\text{VAR}[X]$

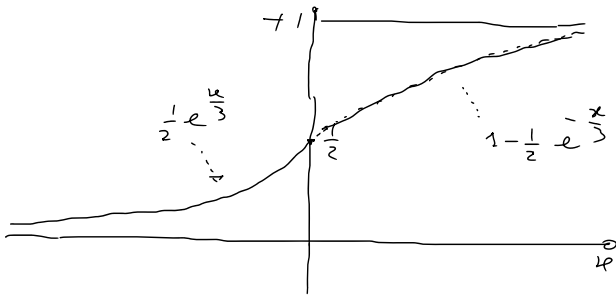
$$f_{X_0}(x) = \frac{1}{2} e^{-|x|} \quad \forall x \in \mathbb{R}$$

$$Y = aX$$

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y}{a}\right) = \frac{1}{3} f_{X_0}\left(\frac{y}{3}\right)$$

$$Y = 3X_0$$

$$F_X(x) = \int_{-\infty}^x f_X(z) dz = \int_{-\infty}^x \frac{1}{6} e^{-\frac{|z|}{3}} dz = \begin{cases} \int_{-\infty}^x \frac{1}{6} e^{\frac{z}{3}} dz = \left[\frac{1}{2} e^{\frac{z}{3}} \right]_{-\infty}^x = \frac{1}{2} e^{\frac{x}{3}} & x < 0 \\ \int_{-\infty}^0 f_X(z) dz + \int_0^x f_X(z) dz = \frac{1}{2} + \int_0^x \frac{1}{6} e^{-\frac{z}{3}} dz \\ = \frac{1}{2} + \left[-\frac{1}{2} e^{-\frac{z}{3}} \right]_0^x = \frac{1}{2} + \frac{1}{2} - \frac{1}{2} e^{-\frac{x}{3}} = 1 - \frac{1}{2} e^{-\frac{x}{3}} & x \geq 0 \end{cases}$$



$$\int_{-\infty}^0 \frac{1}{6} e^{\frac{z}{3}} dz = \left[\frac{1}{2} e^{\frac{z}{3}} \right]_{-\infty}^0 = \frac{1}{2}$$

$$f_X(a+x) = f_X(a-x) \quad \forall x \in \mathbb{R} \Rightarrow \underline{E[X] = a}$$

$$E[X] = \int_{-\infty}^{+\infty} x f_X(x) dx \quad \begin{matrix} \text{disper.} \\ \text{disper.} \end{matrix} \quad \begin{matrix} \text{disper.} \\ \text{disper.} \end{matrix}$$

$$\int_0^{+\infty} x \frac{1}{6} e^{-\frac{x}{3}} dx = \frac{1}{2} \left\{ \left[x e^{-\frac{x}{3}} \right]_0^{+\infty} - \int_0^{+\infty} (-e^{-\frac{x}{3}}) dx \right\} = \frac{1}{2} \left[-3 e^{-\frac{x}{3}} \right]_0^{+\infty} = \frac{3}{2}$$

$$\int_0^{\infty} u \frac{1}{6} e^{-\frac{u}{3}} du = \frac{1}{2} \left\{ \left[u \cdot e^{-\frac{u}{3}} \right]_0^{\infty} - \int_0^{\infty} \left(-e^{-\frac{u}{3}} \right) du \right\} = \frac{1}{2} \left[-3 e^{-\frac{u}{3}} \right]_{u=0}^{\infty} = \frac{3}{2}$$

$$\int_{-\infty}^0 u \frac{1}{6} e^{\frac{u}{3}} du = \frac{1}{2} \left\{ \left[u \cdot e^{\frac{u}{3}} \right]_{-\infty}^0 - \int_{-\infty}^0 e^{\frac{u}{3}} du \right\} = \frac{1}{2} \left\{ - \left[3 e^{\frac{u}{3}} \right]_{-\infty}^0 \right\} = -\frac{3}{2}$$

$$\begin{aligned} \text{VAR}[X] &= E[X^2] = \int_{-\infty}^{\infty} u^2 f_X(u) du = 2 \int_0^{\infty} u^2 f_X(u) du = 2 \int_0^{\infty} u^2 \frac{1}{6} e^{-\frac{u}{3}} du \\ &= \left[u^2 \left(-e^{-\frac{u}{3}} \right) \right]_0^{\infty} - \int_0^{\infty} 2u \left(-e^{-\frac{u}{3}} \right) du = 2 \int_0^{\infty} u e^{-\frac{u}{3}} du = 2 \cdot 9 \left(\frac{3}{2} + 6 \right) = 18 \end{aligned}$$

$$V = \begin{cases} 1 & Z > \eta \\ -1 & Z \leq \eta \end{cases} \quad P[V=1] = P[Z > \eta] = 1 - P[Z \leq \eta] = 1 - F_Z(\eta)$$

$$\begin{aligned} P[V=-1] &= 1 - P[V=1] = 1 - \frac{1}{2} e^{-\frac{\eta}{3}} \\ &= 1 - \left(1 - \frac{1}{2} e^{-\frac{\eta}{3}} \right) \\ &= \frac{1}{2} e^{-\frac{\eta}{3}} \end{aligned}$$

$$\begin{aligned} E[V] &= (+1) P[V=1] + (-1) P[V=-1] = \\ &= \frac{1}{2} e^{-\frac{\eta}{3}} - \left(1 - \frac{1}{2} e^{-\frac{\eta}{3}} \right) = e^{-\frac{\eta}{3}} - 1 \end{aligned}$$

$$E[V^2] = (+1)^2 P[V=1] + (-1)^2 P[V=-1] = P[V=1] + P[V=-1] = 1$$

$$\sigma_V^2 = 1 - \left(1 - e^{-\frac{\eta}{3}} \right)^2 = 2 e^{-\frac{\eta}{3}} - e^{-\frac{2}{3}\eta}$$

Febbraio 2026

X e Y indep. ed. esponential.

$$Z = B(1-X) + (1-B)(Y+2)$$

$$E[X] = 1 \quad E[Y] = \frac{1}{2}$$

$$B \in \{0,1\}, \text{ indep. da } X \text{ e da } Y, \quad P(B=1) = \frac{1}{2}$$

1) Alfabeta, PMF e media di Z

$$2) E[Z | Z > 2]$$

Marzo 2026

$$\text{Gm. Febbraio, ecc. } Z = BX + (1-B)Y$$

$$\phi_Z(s) = E \left[e^{-sZ} \right] \quad s \geq 0$$

$$1) \text{ Trovare } \phi_Z(s)$$

$$2) \text{ Trovare } \phi_Z'(0) \text{ e } \frac{d}{ds} \phi_Z(s) \Big|_{s=0}$$

...

2) trovare $\phi_z(s)$ e $\frac{d}{ds} \phi_z(s) \big|_{s=0}$

3) Definizione $C(B) = E[e^{-Z} | B]$

$f_n \rightarrow P$

$X_n(\omega)$ stocastico a processo discreto

$X_{-1,0}, X_{-1,1}, X_{-1,2}, \dots, X_{1,0}(\omega_1)$

$f_n \rightarrow P$

v.e.

$X_1, X_2, \dots, X_n$

1) Convergenza nel senso delle distribuzioni.

2) Convergenza in prob. forte: $P[|f_n - P| > \varepsilon] \xrightarrow{n \rightarrow \infty} 0$

3) Convergenza in media quadratica $E[|f_n - P|^2] \xrightarrow{n \rightarrow \infty} 0$

4) Convergenza almost sure a.s. $P\left[\lim_{n \rightarrow \infty} f_n = P\right] = 1$

